

Procedure

1. First, a synthetic dataset is created by generating two different groups of data points using Gaussian distributions. These two groups are then combined into a single dataset.
2. Next, the initial parameters of the Gaussian Mixture Model are set. This includes selecting the number of clusters, assigning initial mean values, covariance matrices, and mixing coefficients.
3. A Gaussian density function is defined to calculate the probability of each data point belonging to each cluster.
4. In the Expectation (E) Step, the probability of every data point belonging to each cluster is calculated. These probabilities are called responsibility values.
5. In the Maximization (M) Step, the model parameters such as the mean, covariance matrix, and mixing coefficients are updated based on the responsibility values obtained in the E-Step.
6. The log-likelihood value is then calculated to measure how well the model fits the given dataset after each iteration.
7. The E-Step and M-Step are repeated until the log-likelihood value becomes stable or the maximum number of iterations is reached.
8. After convergence, each data point is assigned to the cluster with the highest probability.
9. Finally, the results are visualized by plotting the generated dataset, the final clustered data with cluster centers, and the log-likelihood convergence graph.